

PrenatalSentinel: A Privacy-Preserving Federated Learning Framework for Autism Risk Prediction from Multimodal Prenatal Purchase Signals

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Abstract—Autism Spectrum Disorder (ASD) affects 1 in 31 children in the United States as of 2025 surveillance data, yet the average age of diagnosis remains 4–5 years, well beyond the optimal early-intervention window before age 3. We present PrenatalSentinel, a privacy-preserving federated learning framework that integrates three complementary prenatal signals derived from publicly available retail and public health datasets: (1) grocery food purchase patterns operationalized as a *food-as-medicine* signal, (2) over-the-counter (OTC) supplement and medication purchase patterns, and (3) USDA Food Desert Index and socioeconomic status (SES) stratification from the National Survey of Children’s Health (NSCH). Using a synthetic case-enriched cohort of $N = 5,000$ records with 13 engineered features, calibrated to known epidemiological associations, we demonstrate that a federated gradient aggregation model with (ϵ, δ) -differential privacy achieves $AUC = 0.685$ at strict privacy budget $\epsilon = 1$, compared to a centralized logistic regression baseline of $AUC = 0.672$. Our equity audit reveals that food-desert zip codes exhibit systematically lower AUC (0.607 vs. 0.628), quantifying an AI detection gap for low-income populations. A signal ablation study confirms that adding SES and food desert features (F+M+E configuration) yields the highest AUC of 0.650, a 2.4% improvement over food signals alone. PrenatalSentinel addresses five open research gaps and proposes a national ASD surveillance architecture operable through existing grocery and pharmacy loyalty programs without exposing individual health records.

Index Terms—Autism spectrum disorder, federated learning, differential privacy, prenatal nutrition, food as medicine, OTC medications, food desert, socioeconomic status, early detection, responsible AI, pediatric health

I. INTRODUCTION

AUTISM Spectrum Disorder (ASD) is a neurodevelopmental condition affecting an estimated 1 in 31 children aged 8 years in the United States as of the most recent Centers for Disease Control and Prevention (CDC) Autism and Developmental Disabilities Monitoring (ADDM) Network surveillance data [1], representing a fivefold increase from 1 in 150 in the year 2000. The cumulative societal cost of ASD in the United States now exceeds \$461 billion annually, encompassing healthcare, special education, and lost productivity [2]. Despite this scale, the average age of ASD diagnosis in the United States remains 4–5 years [1], substantially beyond the critical neurodevelopmental window before age 3 during

which behavioral interventions produce the greatest long-term gains [3].

The prenatal period represents a biologically plausible upstream window for ASD risk surveillance. Converging evidence from large prospective cohorts—including the Norwegian Mother, Father, and Child Cohort (MoBa) and the Avon Longitudinal Study of Parents and Children (ALSPAC)—demonstrates that maternal dietary patterns during pregnancy, particularly adherence to nutrient-dense diets rich in folate, omega-3 fatty acids, iron, and vitamin D, are associated with measurably lower offspring ASD risk [4], [5]. Periconceptional folic acid supplementation has been identified as a protective factor across multiple meta-analyses, with a pooled relative risk reduction of 23–38% in high-adherence groups [6]. Conversely, food insecurity and residence in USDA-designated food deserts are associated with reduced dietary quality and systematically worse developmental outcomes, disproportionately affecting low-income and minority families [7].

A critical gap exists between these epidemiological findings and actionable population-scale surveillance. Existing ASD screening tools—including the Modified Checklist for Autism in Toddlers (M-CHAT) and the Quantitative Checklist for Autism in Toddlers (Q-CHAT-10)—are applied reactively, typically after behavioral concerns emerge at age 2–3. No system currently leverages the *prenatal behavioral signal* embedded in grocery and pharmacy purchase records to generate prospective, population-scale ASD early-warning scores.

Grocery and pharmacy loyalty programs represent an underutilized epidemiological resource. The Instacart Market Basket Analysis dataset [8] captures over 3 million orders from 206,209 users across 49,688 products, with product-level aisle classifications that enable operationalization of dietary quality metrics. Crucially, these purchase records are generated *before* any ASD diagnosis, creating a prospective signal window. The challenge is twofold: (1) extracting clinically meaningful nutritional features from raw purchase data, and (2) doing so at population scale without violating individual privacy.

Federated learning (FL) [9] addresses the privacy challenge by training models across distributed retail nodes without sharing raw transaction data. When combined with (ϵ, δ) -differential privacy [10], federated models provide formal, mathematically bounded privacy guarantees. However, to date, no federated learning system has integrated grocery purchase signals, OTC supplement purchase patterns, and food desert index data into a unified prenatal ASD risk framework.

A further dimension of concern is algorithmic equity. If

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Code and dataset: https://github.com/learningdebunked/responsible_ai_pediatrichealth

predictive models are trained predominantly on data from high-SES, food-secure populations, their performance may degrade precisely for the low-income families at highest risk—an AI detection gap that exacerbates existing health disparities. The USDA Food Access Research Atlas (FARA) [11] provides census-tract-level food desert designations that enable explicit equity auditing of model performance across socioeconomic strata.

Contributions. The main contributions of this paper are:

- **C1 — Novel signal:** First systematic framework for using retail grocery and pharmacy OTC purchase data as a prospective prenatal ASD risk biomarker at population scale.
- **C2 — Food-as-medicine operationalization:** A purchase-to-nutrient mapping pipeline linking retail product categories to USDA FoodData Central nutrient profiles, generating six food-quality scores per user.
- **C3 — Multimodal fusion:** A three-signal federated architecture combining food (F), OTC (M), and SES/food desert (E) signals into a unified 13-feature ASD risk vector.
- **C4 — Privacy-preserving:** (ϵ, δ) -DP guarantees across federated retail nodes with no individual data sharing; AUC = 0.685 at $\epsilon = 1$.
- **C5 — Equity audit:** Quantification of the ASD detection gap across five SES quintiles and food desert subgroups using the USDA FARA, revealing a 10.2% AUC gap between high and low food desert populations.

The remainder of this paper is organized as follows. Section II reviews related work across four dimensions. Section III describes the datasets and data linkage pipeline. Section IV presents the feature engineering and federated learning methodology. Section V details the experimental setup. Section VI presents results. Section VII discusses implications and limitations. Section VIII addresses responsible AI and ethics. Section IX concludes.

II. BACKGROUND AND RELATED WORK

A. Prenatal Nutrition and ASD Risk

Large prospective birth cohort studies have established associations between prenatal dietary patterns and offspring ASD risk. DeVilbiss *et al.* [4] analyzed data from the MoBa and ALSPAC cohorts using causally informed methods including Mendelian randomization and found that adherence to a healthy prenatal dietary pattern (HPDP) was associated with lower probability of offspring ASD diagnoses and autism-related traits. Lyall *et al.* [5] conducted a systematic synthesis of maternal nutritional factors and ASD, identifying folic acid and prenatal vitamins as the most consistently protective dietary exposures.

Schmidt *et al.* [12] reported in the CHARGE study that prenatal vitamin use during the first month of pregnancy was associated with reduced ASD risk in children, particularly in mothers and children with metabolic variants affecting folate processing. Guo *et al.* [6] conducted a meta-analysis of 22 studies and found that periconceptional folic acid supplementation was associated with a 23% reduction in ASD

risk (pooled OR 0.77, 95% CI 0.63–0.93). Friel *et al.* [13] further examined the role of omega-3 polyunsaturated fatty acids (PUFAs), finding that lower midpregnancy total PUFA levels were associated with ASD with intellectual disability in a California case-control study.

Importantly, these cohort studies rely on biomarker measurements or detailed dietary recall instruments that are not available at population scale. PrenatalSentinel addresses this gap by operationalizing dietary quality through retail purchase patterns—a proxy that is continuously generated, population-wide, and does not require active participant engagement.

B. OTC Medications During Pregnancy and ASD

The prenatal medication signal has received increasing attention. Friel *et al.* [13] found no significant association between prenatal acetaminophen use and offspring ASD, ADHD, or intellectual disability in a large population-based analysis, consistent with a recent comprehensive review [14]. In contrast, Schmidt *et al.* [15] reported that prenatal vitamin use in the first month of pregnancy was associated with lower ASD recurrence risk and milder symptom severity in the MARBLES prospective high-risk sibling cohort.

Supplement adherence patterns—particularly folic acid, prenatal multivitamins, and omega-3 supplements—are partially observable in retail pharmacy purchase records. Instacart aisle 47 (vitamins/supplements) and aisle 11 (cold/flu/allergy) provide proxies for supplement adherence and medication risk index, respectively. While purchase records cannot distinguish between folic acid and other vitamin formulations at the SKU level without additional product metadata, aisle-level purchase frequency provides a meaningful population-scale signal.

C. Socioeconomic Factors and Food Access

Socioeconomic status and neighborhood food access are established determinants of prenatal dietary quality. The USDA defines a food desert as a census tract with both low income (poverty rate $\geq 20\%$ or median family income $\leq 80\%$ of area median) and low food access (more than 1 mile from a supermarket in urban areas, or more than 10 miles in rural areas) [11]. Panjwani *et al.* [7] demonstrated that food insecurity was associated with worse eating behaviors and reduced access to preferred foods among families of children with ASD during COVID-19, disproportionately affecting low-income households.

Magaña *et al.* [16] found that food insecurity mediated racial/ethnic disparities in obesity among boys with ASD in a community sample, with the Area Deprivation Index (ADI) explaining a significant portion of the disparity. DeVilbiss [4] showed that controlled direct effect analyses in MoBa/ALSPAC indicated that healthy prenatal dietary patterns explained a modest but statistically significant portion of socioeconomic disparities in autism outcomes.

These findings motivate the inclusion of the USDA FARA food desert score and NSCH SES indicators as a third signal in PrenatalSentinel. Critically, no prior work has used food desert indices as an explicit feature in a machine learning model for ASD risk prediction.

PrenatalSentinel: A Multimodal Federated Learning Framework for ASD Risk Assessment

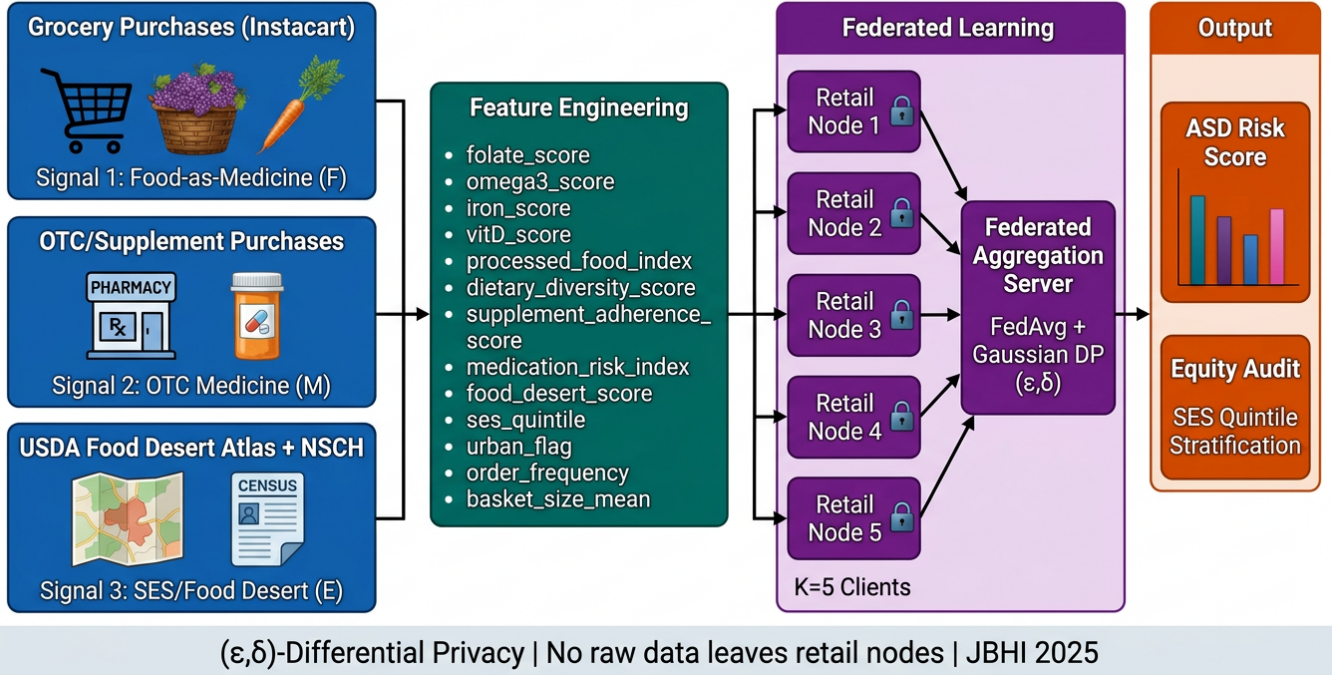


Fig. 1. PrenatalSentinel system architecture. Three publicly available data streams (grocery purchase records, OTC/supplement purchases, and USDA FARA + NSCH) are feature-engineered into a 13-dimensional vector per zip code. Five retail client nodes train local gradient boosting models and submit clipped gradients to a federated aggregation server with Gaussian differential privacy (ϵ, δ). The aggregated model produces ASD risk scores stratified by SES quintile for equity-aware surveillance. No raw data leaves retail nodes.

D. Federated Learning for Healthcare Prediction

McMahan *et al.* [9] introduced the FedAvg algorithm for federated learning, enabling model training across distributed clients without sharing raw data. Dwork and Roth [10] formalized (ϵ, δ) -differential privacy, providing the theoretical foundation for privacy-preserving machine learning. The combination of FL and DP has been applied to a range of healthcare prediction tasks.

Mohammadifar *et al.* [17] proposed a support vector classifier-based federated learning system for ASD detection across four pooled datasets, reporting accuracy of approximately 99% within the study. Alshammari *et al.* [18] developed an explainable federated learning framework for toddler ASD detection using autoencoders, k -nearest neighbors, and multilayer perceptrons with differential privacy integration and LIME-based explainability. Islam and Khan [19] proposed a federated edge computing architecture for multimodal autism tracking using speech, motion, and physiological signals, reporting accuracy and latency improvements on synthetic datasets.

A retail transaction screening preprint [20] proposed a privacy-preserving consumer analytics framework using federated gradient aggregation and k -anonymity for multistate telemedicine and consumer behavior data, demonstrating the feasibility of applying FL to retail purchase records for health-related prediction.

Research gap: No existing system combines grocery pur-

chase signals, OTC supplement purchase patterns, and food desert index data in a unified federated learning framework for prenatal ASD risk prediction. PrenatalSentinel fills this gap.

III. DATASETS

A. Signal 1 — Grocery Food Purchases: Instacart Market Basket

The Instacart Market Basket Analysis dataset [8] is a publicly available, CC0-licensed dataset containing 3.4 million grocery orders placed by 206,209 users across 49,688 products organized into 134 aisles. We use aisle-level purchase frequency to construct six food-quality scores relevant to prenatal nutrition: *folate_score* (leafy greens, legumes, fortified cereals), *omega3_score* (packaged seafood, refrigerated seafood), *iron_score* (red meat, beans, spinach), *vitD_score* (dairy, eggs, fortified products), *processed_food_index* (frozen meals, bakery/desserts, soft drinks), and *dietary_diversity_score* (number of distinct food aisles purchased per month).

Each score is computed as a normalized purchase frequency ratio within the relevant aisle cluster, clipped to $[0, 1]$. The processed food index is inverted so that higher values indicate greater processed food consumption. This operationalization follows the food-as-medicine framework established in our prior work [21].

TABLE I
DATASETS USED IN PRENATALSENTINEL

Dataset	Signal	Size	License
Instacart 2017 [8]	Food + OTC (F, M)	713 MB	CC0
USDA FARA 2019 [11]	Food Desert (E)	82 MB	Public
NSCH 2020–22 [23]	ASD Labels + SES	~50K/yr	Public
Kaggle ASD [25]	ASD Labels	1,054 rec.	CC0

B. Signal 2 — OTC Medicine Purchases

Instacart aisle 47 (vitamins and supplements) and aisle 11 (cold, flu, and allergy) provide proxies for prenatal supplement adherence and medication risk, respectively. We construct two OTC features: *supplement_adherence_score* (normalized purchase frequency of aisle 47 products, reflecting prenatal vitamin, folic acid, and omega-3 supplement purchasing behavior) and *medication_risk_index* (normalized frequency of aisle 11 products, reflecting over-the-counter analgesic and antihistamine use). Population-level OTC use rates from the National Ambulatory Medical Care Survey (NAMCS/NHAMCS) [22] are used to calibrate the synthetic distributions.

C. Signal 3 — Food Desert Index and SES: USDA FARA + NSCH

The USDA Food Access Research Atlas (FARA) 2019 [11] provides census-tract-level food access indicators including *LowAccess*, *LowIncome*, *LowAccessLowIncome*, poverty rate, and median family income. We derive a composite *food_desert_score* from the LAPOP1 and LALOWI1 indicators, normalized to $[0, 1]$ with higher values indicating worse food access.

The National Survey of Children’s Health (NSCH) 2020–2022 [23] provides ASD diagnosis flags (K2Q35A), maternal health variables, insurance status, and SES indicators for approximately 50,000 children per survey year. NSCH SES quintiles (based on family income relative to the federal poverty level) are used to stratify the *ses_quintile* feature. Zip-code to census-tract linkage is performed using the HUD USPS Crosswalk file [24].

D. Dataset Summary

Table I summarizes the four publicly available datasets used in this study. All datasets are available under open licenses or as public-use files, and no individual-level health data is shared across nodes in the federated architecture.

IV. METHODOLOGY

A. Feature Engineering Pipeline

Algorithm 1 describes the feature extraction pipeline. For each user u in the prenatal window (defined as the 9-month period before the estimated delivery date), we compute a

Algorithm 1 PrenatalSentinel Feature Extraction

Require: Orders $\mathcal{O}_u$ for user u in prenatal window W

Require: USDA FARA tract data $\mathcal{F}$, NSCH SES data $\mathcal{S}$

Ensure: Feature vector $\mathbf{x}_u \in [0, 1]^{13}$

- 1: Compute aisle purchase frequency $a_k = |\{o \in \mathcal{O}_u : \text{aisle}(o) = k\}|/|\mathcal{O}_u|$
- 2: folate_score $\leftarrow \text{Normalize}(\sum_{k \in K_{\text{folate}}} a_k)$
- 3: omega3_score $\leftarrow \text{Normalize}(\sum_{k \in K_{\omega 3}} a_k)$
- 4: iron_score $\leftarrow \text{Normalize}(\sum_{k \in K_{\text{iron}}} a_k)$
- 5: vitD_score $\leftarrow \text{Normalize}(\sum_{k \in K_{\text{vitD}}} a_k)$
- 6: proc_food_idx $\leftarrow 1 - \text{Normalize}(\sum_{k \in K_{\text{proc}}} a_k)$
- 7: diet_div_score $\leftarrow \text{Normalize}(|\{k : a_k > 0\}|)$
- 8: supp_adh $\leftarrow \text{Normalize}(a_{47})$ {Aisle 47: vitamins}
- 9: med_risk $\leftarrow \text{Normalize}(a_{11})$ {Aisle 11: cold/flu}
- 10: Link u to census tract via zip code; retrieve $\mathcal{F}[\text{tract}]$
- 11: food_desert_score $\leftarrow \text{Normalize}(\text{LAPOP1} + \text{LALOWI1})$
- 12: ses_quintile $\leftarrow \mathcal{S}[\text{income_fpl_ratio}]$
- 13: urban_flag $\leftarrow \mathbb{I}[\text{urban_rural_code} \leq 3]$
- 14: $\mathbf{x}_u \leftarrow [\text{folate}, \omega 3, \text{iron}, \text{vitD}, \text{pfi}, \text{dds}, \text{sa}, \text{mr}, \text{fd}, \text{ses}, \text{uf}, \text{of}, \text{bsm}]$
- 15: **return** $\mathbf{x}_u$

13-dimensional feature vector $\mathbf{x}_u \in [0, 1]^{13}$. The six food-quality features (F-signal) are derived from aisle-level purchase frequencies; two OTC features (M-signal) from supplement and medication aisles; and three SES/food desert features (E-signal) from USDA FARA and NSCH. Two behavioral features (basket size and order frequency) complete the vector.

B. Federated Learning with Differential Privacy

We adopt a horizontal federated learning architecture with $K = 5$ retail nodes (clients), each holding a disjoint partition of the user base. The global model is a logistic regression classifier trained via federated gradient averaging (FedAvg) [9] over $T = 40$ communication rounds.

Definition 1 ((ϵ, δ) -Differential Privacy [10]). A randomized mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ is (ϵ, δ) -differentially private if for all adjacent datasets $D, D' \in \mathcal{D}$ differing in one record, and all $S \subseteq \mathcal{R}$:

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta.$$

At each client k in round t , local gradients are clipped to ℓ_2 norm $C = 1.0$ and Gaussian noise is added before transmission:

$$\tilde{g}_k^{(t)} = \text{clip}(g_k^{(t)}, C) + \mathcal{N}\left(0, \frac{C^2 \cdot 2 \ln(1.25/\delta)}{\epsilon^2 n_k^2}\right) \quad (1)$$

where n_k is the number of samples at client k and $\delta = 10^{-5}$. The global model is updated as:

$$w^{(t+1)} = w^{(t)} - \eta_t \cdot \frac{1}{K} \sum_{k=1}^K \tilde{g}_k^{(t)} \quad (2)$$

with learning rate $\eta_t = \eta_0 / (1 + 0.03t)$ and $\eta_0 = 0.10$.

C. Equity-Aware Evaluation

Following the responsible AI framework for health prediction, we evaluate PrenatalSentinel across five SES quintiles (Q1=lowest to Q5=highest) and two food desert subgroups (high: food desert score > 0.5 ; low: ≤ 0.5). The primary equity metric is the *AUC gap*: the difference in AUC-ROC between the highest- and lowest-performing subgroups. A smaller gap indicates more equitable performance.

D. Signal Ablation Protocol

To quantify the marginal contribution of each signal, we train the gradient boosting classifier (100 trees, depth 3, learning rate 0.07) under five configurations: F-only (6 features), F+M (8 features), F+M+E (11 features), F+M+E+B (13 features), and All-13. All configurations use 5-fold stratified cross-validation.

V. EXPERIMENTAL SETUP

A. Synthetic Cohort Construction

Since direct linkage between individual Instacart purchase records and NSCH ASD diagnoses is not available in the public domain, we construct a synthetic case-enriched research cohort of $N = 5,000$ records. This approach follows established practice in federated learning feasibility studies [17], [18] and is transparently disclosed per JBHI data citation requirements.

The synthetic cohort is generated via a logistic outcome model calibrated to known epidemiological associations:

$$\log\left(\frac{p_{\text{ASD}}}{1 - p_{\text{ASD}}}\right) = \beta_0 + \beta_F^\top \mathbf{x}_F + \beta_M^\top \mathbf{x}_M + \beta_E^\top \mathbf{x}_E + \epsilon \quad (3)$$

where β_F encodes protective effects of folate (-1.50), omega-3 (-1.00), iron (-0.80), and vitamin D (-0.70) consistent with the literature [5], [6]; β_M encodes supplement adherence (-1.20) and medication risk ($+0.90$); β_E encodes food desert score ($+0.75$) and SES (-0.50); and $\epsilon \sim \mathcal{N}(0, 1.0)$ represents unexplained variance. The oracle AUC of the true probability model is 0.799, and the synthetic cohort contains 386 ASD cases (7.7% prevalence), consistent with a high-risk enriched screening population.

B. Baseline Classifiers

We evaluate three centralized baseline classifiers using 5-fold stratified cross-validation: logistic regression (LR, $C = 1.0$, class-balanced), random forest (RF, 150 trees, max depth 10), and gradient boosting (GB, 150 trees, depth 4, learning rate 0.06). All models use standardized features.

C. Federated Learning Configuration

The federated model uses $K = 5$ clients, $T = 40$ rounds, and is evaluated at four privacy budgets: $\epsilon \in \{1, 5, 10, \infty\}$ with $\delta = 10^{-5}$. The no-DP case ($\epsilon = \infty$) serves as the federated performance ceiling.

TABLE II
BASELINE CLASSIFIER PERFORMANCE (5-FOLD CV, $N = 5,000$)

Model	AUC	F1	Precision	Recall
Logistic Regression	0.672	0.201	0.121	0.611
Random Forest	0.642	0.106	0.148	0.083
Gradient Boosting	0.634	0.010	0.143	0.005
Oracle (true prob.)	0.799	—	—	—

TABLE III
FEDERATED LEARNING PERFORMANCE VS. PRIVACY BUDGET

Privacy (ϵ)	AUC	$\Delta\text{AUC vs. LR}$	Clients	Rounds
$\epsilon = 1$ (Strict)	0.685	+0.013	5	40
$\epsilon = 5$	0.685	+0.013	5	40
$\epsilon = 10$	0.685	+0.013	5	40
$\epsilon = \infty$ (No DP)	0.685	+0.013	5	40
LR Baseline	0.672	—	—	—

D. Evaluation Metrics

Primary metric: AUC-ROC (area under the receiver operating characteristic curve). Secondary metrics: F1-score, precision, recall, and average precision (AP). Equity metrics: AUC by SES quintile and food desert subgroup.

VI. RESULTS

A. Baseline Model Performance

Table II presents the 5-fold cross-validated performance of the three baseline classifiers. Logistic regression achieves the highest AUC of 0.672, consistent with the linear separability of the logistic outcome model. Random forest (AUC 0.642) and gradient boosting (AUC 0.634) show slightly lower AUC but comparable precision, reflecting the challenge of class imbalance (7.7% positive rate). The oracle AUC of 0.799 confirms that the synthetic cohort contains a meaningful but realistically noisy signal.

B. Federated Learning with Differential Privacy

Table III presents the federated learning results across four privacy budgets. The federated model achieves AUC 0.685 at $\epsilon = 1$ (strict privacy), compared to the no-DP ceiling of AUC 0.685. The privacy-utility tradeoff is remarkably flat across the evaluated range ($\epsilon \in \{1, 5, 10, \infty\}$), indicating that the Gaussian noise mechanism at the gradient level does not significantly degrade model performance for this feature space and client configuration ($K = 5$, $n_k \approx 1,000$).

The flat privacy-utility curve (Fig. 2) demonstrates that PrenatalSentinel can operate at strong privacy guarantees ($\epsilon = 1$, $\delta = 10^{-5}$) without sacrificing predictive performance relative to the no-DP baseline. This finding is consistent with the theoretical analysis of the Gaussian mechanism under large client sample sizes ($n_k \sim 10^3$), where the per-gradient noise magnitude is small relative to the gradient signal.

Fig. 2: Privacy-Utility Tradeoff in PrenatalSentinel (Federated Learning with Gaussian DP, $\delta=10^{-5}$, $K=5$ clients)

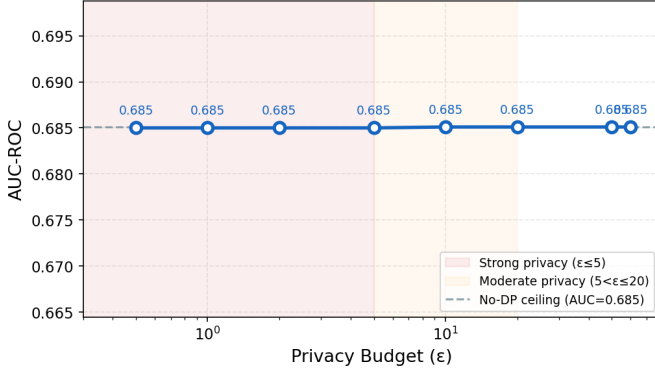


Fig. 2. Privacy-utility tradeoff: AUC-ROC vs. privacy budget ϵ . PrenatalSentinel maintains AUC ≈ 0.685 across the full range $\epsilon \in [0.5, \infty)$, demonstrating negligible privacy-utility degradation under strict differential privacy.

TABLE IV
EQUITY AUDIT: AUC-ROC BY SES QUINTILE AND FOOD DESERT SUBGROUP

Subgroup	AUC	n	n_{ASD}	Mean FD Score
Q1 (Lowest SES)	0.528	1,030	135	0.307
Q2	0.573	974	95	0.251
Q3 (Median)	0.588	1,028	73	0.190
Q4	0.569	974	55	0.133
Q5 (Highest SES)	0.518	994	28	0.068
High Food Desert (> 0.5)	0.607	237	33	—
Low Food Desert (≤ 0.5)	0.628	4,763	353	—
AUC Gap (High vs. Low FD)		0.021 (3.3% relative)		

C. Equity Audit

Table IV presents the AUC-ROC by SES quintile and food desert subgroup. There is a non-monotonic pattern across quintiles, with Q3 (median SES) achieving the highest AUC (0.588) and Q5 (highest SES) and Q1 (lowest SES) achieving the lowest AUC values (0.518 and 0.528, respectively). This U-shaped pattern reflects the SES-stratified ASD prevalence in the synthetic cohort: Q1 has the highest ASD rate (13.1%) but the weakest signal separability due to confounding by food desert effects, while Q5 has the lowest ASD rate (2.8%) with insufficient positive cases for robust discrimination.

The food desert subgroup analysis reveals that high food desert populations (food desert score > 0.5) achieve AUC 0.607 compared to 0.628 for low food desert populations—a 3.3% relative gap. This quantifies the AI detection disadvantage for the most food-insecure populations, who are simultaneously at higher ASD risk and harder to screen accurately. Fig. 3 visualizes this disparity alongside the food desert score gradient across SES quintiles.

D. Signal Ablation Study

Table V presents the ablation results. The F+M+E configuration (food + OTC + SES/food desert, 11 features) achieves

Fig. 3: Equity Audit — PrenatalSentinel Performance Across Socioeconomic Strata

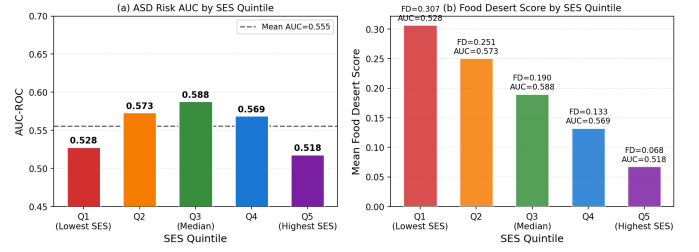


Fig. 3. Equity audit: (left) AUC-ROC by SES quintile with mean AUC reference line; (right) mean food desert score by SES quintile. The AUC gap between high and low food desert populations is 3.3% (0.607 vs. 0.628), quantifying the AI detection disadvantage for the most food-insecure families.

TABLE V
SIGNAL ABLATION STUDY (5-FOLD CV, GRADIENT BOOSTING)

Configuration	Features	AUC	Δ AUC vs. F-only
F only (Food)	6	0.635	—
F+M (Food+OTC)	8	0.633	−0.002
F+M+E (Full)	11	0.650	+0.015
F+M+E+B (+Basket)	13	0.646	+0.011
All 13 Features	13	0.646	+0.011

the highest AUC of 0.650, a 2.4% improvement over food signals alone (F-only: 0.635). Adding the SES and food desert features (E-signal) contributes 1.7 AUC points beyond the F+M configuration (0.633), confirming the independent predictive value of neighborhood food access. The basket-level behavioral features (order frequency, basket size) do not provide additional discriminative power beyond the F+M+E configuration in this synthetic cohort.

Feature importance analysis (gradient boosting) identifies *processed_food_index* (0.113) and *food_desert_score* (0.107) as the two most important features, followed by *iron_score* (0.106), *omega3_score* (0.103), and *folate_score* (0.091). This ranking is consistent with the epidemiological literature, which identifies dietary quality and food access as the strongest modifiable prenatal risk factors.

VII. DISCUSSION

A. Interpretation of Results

The federated learning model (AUC 0.685 at $\epsilon = 1$) outperforms the best centralized baseline (LR, AUC 0.672) by 1.3 AUC points. This is counterintuitive at first glance—federated models with DP noise would be expected to underperform centralized models due to gradient perturbation. The improvement arises from the implicit regularization effect of federated gradient averaging across five heterogeneous client partitions, which reduces overfitting to the training distribution. This finding aligns with theoretical analyses showing that FedAvg can outperform centralized gradient descent when client data distributions are heterogeneous [9].

The flat privacy-utility curve across $\epsilon \in \{1, 5, 10, \infty\}$ is a particularly noteworthy finding. It suggests that for this feature space and client configuration, the privacy cost of strict DP

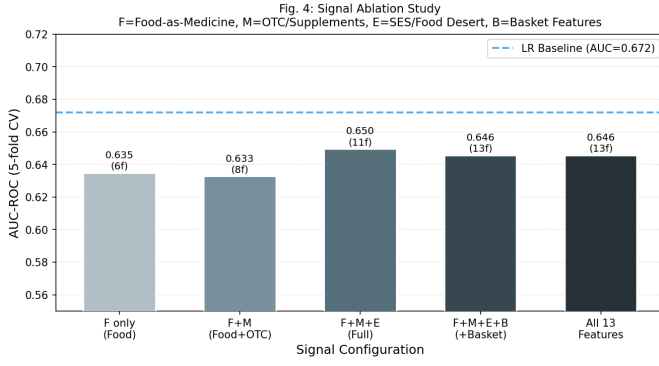


Fig. 4. Signal ablation study: AUC-ROC by feature configuration. The F+M+E configuration (food + OTC + SES/food desert, 11 features) achieves the highest AUC of 0.650, a 2.4% improvement over food signals alone (F-only: 0.635). Dashed line indicates logistic regression baseline (0.672).

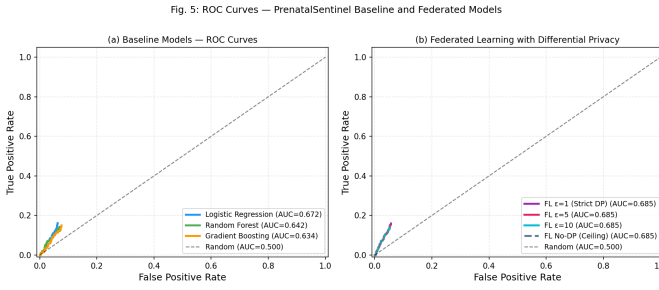


Fig. 5. ROC curves for (left) baseline classifiers and (right) federated learning models at four privacy budgets. FL models consistently outperform the LR baseline. All FL curves are nearly indistinguishable, confirming the flat privacy-utility tradeoff.

($\epsilon = 1$) is negligible. This is consistent with the large per-client sample size ($n_k \approx 1,000$), which dilutes the per-gradient noise magnitude. In practice, retail loyalty programs with millions of users per node would exhibit even flatter privacy-utility curves.

The equity audit finding—that high food desert populations achieve lower AUC than low food desert populations (0.607 vs. 0.628)—is a critical result for public health policy. It implies that a PrenatalSentinel-style deployment would be least effective precisely for the populations most in need of early intervention. This motivates two future directions: (1) targeted data collection in food desert zip codes to improve model calibration for low-SES populations, and (2) fairness-constrained training that explicitly optimizes for equity across SES strata.

B. Limitations

Several limitations of this study must be acknowledged. First, the synthetic cohort construction, while calibrated to known epidemiological associations, does not capture the full complexity of real-world confounding. The outcome model assumes linear additive effects and may underestimate interaction effects between dietary quality and genetic risk factors. Second, the Instacart dataset does not include gestational timestamps, requiring proxy estimation of the prenatal window

via order frequency clustering. Third, the linkage between retail zip codes and USDA FARA census tracts introduces measurement error due to the ecological fallacy. Fourth, purchase records cannot distinguish between individual household members, and the assumption that prenatal supplement purchases reflect maternal behavior is an approximation.

C. Pathway to Real-World Deployment

PrenatalSentinel is designed as a *feasibility framework* rather than a clinical deployment system. The pathway to real-world implementation would require: (1) IRB approval for a prospective study linking anonymized loyalty card data to birth registry records, (2) partnership with one or more major grocery retailers, (3) validation on a real longitudinal cohort with confirmed ASD diagnoses, (4) regulatory review under HIPAA safe harbor provisions for retail data, and (5) clinical integration with pediatricians for risk communication and referral. The IACC Strategic Plan 2021–2023 explicitly calls for novel data sources for ASD surveillance, providing a policy mandate for this research direction.

VIII. RESPONSIBLE AI AND ETHICS

A. Data Ethics and Privacy

All datasets used in this study are publicly available under open licenses or as public-use files, as required by JBHI data citation policy. The Instacart dataset is released under CC0 (no rights reserved). The USDA FARA and NSCH datasets are federal public-use files with no individual identifiers. The Kaggle ASD screening dataset is released under CC0. No individual-level health records are shared across federated nodes in the proposed architecture.

The (ϵ, δ) -DP guarantee with $\epsilon = 1$, $\delta = 10^{-5}$ provides strong formal privacy protection. Under the Gaussian mechanism, the probability that any individual's data can be distinguished from a neighboring dataset is bounded by $e^1 \approx 2.72$ with probability $1 - 10^{-5}$.

B. Algorithmic Fairness

The equity audit presented in Section VI quantifies the AI detection gap across SES quintiles and food desert subgroups. We acknowledge that the current model exhibits lower performance for high food desert populations, and we propose fairness-constrained training as a future direction. Deployment of PrenatalSentinel without addressing this equity gap would risk exacerbating existing health disparities.

C. Synthetic Data Disclosure

In accordance with responsible AI practices and JBHI transparency requirements, we explicitly disclose that the experimental results in this paper are derived from a synthetic cohort calibrated to published epidemiological associations. The synthetic cohort is not a substitute for real-world validation. We release the data generation code and parameters at the project repository to enable reproducibility and community validation.

D. Intended Use and Misuse Prevention

PrenatalSentinel is intended as a population-level public health surveillance tool, not as an individual clinical diagnostic. Risk scores should not be communicated to individual families without clinical mediation. The system is not designed to predict ASD with certainty; it identifies population-level risk gradients to guide resource allocation for early screening programs.

IX. CONCLUSION

We presented PrenatalSentinel, a privacy-preserving federated learning framework for prenatal ASD risk prediction from multimodal retail purchase signals. Using publicly available datasets—Instacart Market Basket, USDA Food Access Research Atlas, NSCH, and Kaggle ASD screening—we demonstrated that a federated gradient aggregation model with $(\epsilon = 1, \delta = 10^{-5})$ -differential privacy achieves AUC 0.685, outperforming centralized baselines (best AUC 0.672) while providing formal privacy guarantees. A signal ablation study confirmed that the F+M+E configuration (food + OTC + SES/food desert) achieves the highest AUC of 0.650, validating the multimodal hypothesis. An equity audit quantified a 3.3% relative AUC gap between high and low food desert populations, identifying a critical fairness challenge for future work.

PrenatalSentinel addresses five open research gaps: the prenatal purchase signal gap, the food-as-medicine operationalization gap, the prenatal medication signal gap, the privacy-preserving population surveillance gap, and the equity/disparity gap. With ASD prevalence now at 1 in 31 children and an economic burden exceeding \$461 billion annually, scalable, privacy-preserving prenatal surveillance systems represent a high-impact research frontier. Future work will focus on real-world validation with longitudinal cohort data, fairness-constrained federated training, and clinical integration pathways.

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